

Chit 4 — MySQL: Instructor/Student/Course Queries

Table Creation & Sample Data

```
CREATE TABLE student (  
  S_ID INT PRIMARY KEY, name VARCHAR(50),  
  dept_name VARCHAR(50), tot_cred INT  
);  
CREATE TABLE instructor (  
  T_ID INT PRIMARY KEY, name VARCHAR(50),  
  dept_name VARCHAR(50), salary DECIMAL(10,2)  
);  
CREATE TABLE course (  
  course_id VARCHAR(10) PRIMARY KEY, title VARCHAR(100),  
  dept_name VARCHAR(50), credits INT  
);  
  
INSERT INTO instructor VALUES  
  (1, 'Amol', 'Computer', 45000), (2, 'Amit', 'Computer', 40000),  
  (3, 'Priya', 'Science', 38000), (4, 'Ravi', 'Maths', 50000),  
  (5, 'Neha', 'Computer', 48000);  
  
INSERT INTO student VALUES  
  (1, 'Samantha', 'Computer', 60), (2, 'Ramesh', 'Science', 45),  
  (3, 'Ameya', 'Computer', 70), (4, 'Kamal', 'Maths', 55);  
  
INSERT INTO course VALUES  
  ('CS101', 'DBMS', 'Computer', 4), ('CS102', 'OS', 'Computer', 3),  
  ('SC101', 'Physics', 'Science', 3);
```

i. Average salary of departments where avg salary > 42000

```
SELECT dept_name, AVG(salary) AS avg_salary  
FROM instructor  
GROUP BY dept_name  
HAVING AVG(salary) > 42000;
```

ii. Increase salary of Computer department instructors by 10%

```
UPDATE instructor  
SET salary = salary * 1.10  
WHERE dept_name = 'Computer';
```

iii. Find instructors whose name is neither 'Amol' nor 'Amit'

```
SELECT name FROM instructor  
WHERE name NOT IN ('Amol', 'Amit');
```

iv. Find student names containing 'am' as substring

```
SELECT name FROM student  
WHERE name LIKE '%am%';
```

v. Names of Computer students who take 'DBMS' courses

```
-- Assumes takes(S_ID, course_id) table exists  
SELECT s.name FROM student s  
JOIN takes t ON s.S_ID = t.S_ID
```

```
JOIN course c ON t.course_id = c.course_id  
WHERE s.dept name = 'Computer' AND c.title = 'DBMS';
```